My PA – Personal Assistant Chatbot

Comprehensive Technical Architecture, Verification, & Multi-Device UI Documentation

Author: Anisa Arum	**Course:** Hacktiv8 AI Bootcamp	**Version:** 1.0 (Production)
Live URL: https://hacktiv-okt-chatbot.vercel.app	**AI Engine:** Gemini 2.5 Flash	**Database:** Neon PostgreSQL + pgvector

1. Executive Overview & Technical Stack

My PA is a stream-of-consciousness personal assistant web application engineered to ingest unstructured natural language thoughts, voice memos, and photos, converting them dynamically into structured database entities (Tasks, Expenses, Groceries, Notes, and Reminders). It includes a 19-skill interactive command engine, multi-device user scoping, 3-tier memory RAG, and automated background SMTP email dispatch.

Layer	Technology & Choice	Key Role & Rationale
Frontend Framework	Next.js 14 (App Router) + TypeScript 5	Server-Side Rendering, API routes, fast client transitions, PWA standalone compatibility.
Styling & UX	Tailwind CSS + Glassmorphism + CSS Variables	Responsive dual design: Desktop sidebar navigation + Mobile bottom app drawer, smooth Dark/Light mode.
AI & Intent Model	Google Gemini 2.5 Flash (`@google/genai`)	Low latency, generous free-tier throughput, JSON Schema structured output guarantee.
Vector RAG Engine	Gemini `text-embedding-004` (768 dimensions)	Powers hybrid semantic vector search over long-term notes and bookmarks.
Cloud Database	Neon Serverless PostgreSQL 16 + `pgvector`	Serverless pooled connection with native pgvector extension for zero-maintenance cloud persistence.
Local Fallback Store	Resilient JSON Store (`my_pa_storage.json`)	Embedded local database engine with zero external setup requirement for offline/dev modes.
Email & Reminders	Nodemailer + Gmail SMTP (App Password)	Direct automated notification dispatch to the user's verified Gmail address.
Hosting & Auth	Vercel Edge Platform + Google OAuth 2.0	Global SSL CDN edge deployment with passwordless Google Identity Services authentication.

2. AI Architecture & Prompt Engineering System Instructions

The model acts as an intelligent router and structured entity extraction compiler. Below is the exact prompt engineering and contextual database injection contract governing Gemini:

Core System Prompt & Processing Contract:

Role: You are My PA (Personal Assistant), a dedicated and highly accurate AI assistant that parses the user's natural thoughts into structured actions and answers questions about their data.

Context Injected per Request:

• Current Local Timestamp & User Timezone (e.g. *Asia/Jakarta*).

• Active Database Snapshot (Filtered tasks, pending expenses, active groceries, rolling conversational summary).

Mode 1: Question / Query Mode:

When the user queries history (*'task ku yang berkaitan sama work apa aja?'*, *'what did I spend today?'*):

1. Answer ACCURATELY based ONLY on the stored database records.
2. Pay strict attention to categories (e.g., if asked for 'work', exclude 'learning' like SAP or personal errands).
3. DO NOT fabricate or create new records. Return empty arrays for entities and output answers in conversational_reply.

Mode 2: Recording / Action Mode:

• **Groceries vs. Expenses Rule:** Food/ingredient items without explicit currency prices MUST be classified as groceries, NEVER expenses.

• **Explicit Prices Rule:** Transactions with numerical money (e.g., *'makan siang 35k'*, *'bayar listrik 150rb'*) create DRAFT expenses.

• **Reminder Date Calculation:** Natural relative phrases (*'besok jam 6 pagi'*) are parsed into strict target ISO timestamps (*YYYY-MM-DDT06:00:00*).

3. Built-in 19 Skills Engine (Slash Commands)

My PA incorporates an instant-response Skills Engine triggered via keyboard slash commands (/) or automated intent detection. Commands such as /info execute in 0ms without consuming AI quota.

Slash Command	Category	Description & Execution Action
/info (or /config)	System	Displays instant cheat sheet of all 19 skills, active AI model, and sliders without quota.
/today	Planning	Compiles today's active tasks, impending deadlines, and today's total spending summary.
/daily-briefing	Planning	Generates a structured morning agenda with priority to-dos and unpaid bills.
/deadline	Tasks	Scans database for upcoming deadlines within N days (default 7 days).
/recap	Review	Provides structured recap of past activity (e.g. */recap yesterday*).
/todo	Tasks	Explicitly creates or marks tasks completed (e.g. */todo finish QPR*).
/expense	Finance	Logs financial expenditure and shows current monthly tally (e.g. */expense dinner 50k*).
/export-expenses	Finance	Generates and downloads a spreadsheet-ready CSV of all recorded transactions.
/grocery	Shopping	Manages shopping checklist items (e.g. */grocery add eggs, olive oil*).
/note	Knowledge	Saves free-form notes with automated tagging and 768-dim vector embeddings.
/web-clipper (/link)	Knowledge	Extracts webpage title and summary, indexing URL into bookmarks.
/plan	Roadmap	Deconstructs long-term objectives into concrete milestones and target dates.
/remind	Alerts	Creates an in-app popup notification at the designated time.
/email-reminder	Alerts	Schedules an alert dispatched to Gmail SMTP inbox at the requested hour.
/send-email	Communication	Summarizes selected records and dispatches a report directly to user's registered email.
/review	Review	Generates weekly productivity audit: completed tasks vs pending, and spending breakdown.
/search	RAG	Performs hybrid semantic cosine-similarity search across all stored notes.
/undo	Recovery	Rolls back and archives the most recent database entry created within the last 10 minutes.
/backup	Security	Generates full JSON database export and sends attachment to user's email instantly.

4. Accuracy Benchmark & Test Suite Results

The system has been evaluated through a dual verification pipeline: **End-to-End System QA Automation** and **Cognitive Domain Accuracy Verification**.

Test Suite & Target Area	Test Cases	Status	Observed Performance & Accuracy
Suite 1: User Scoping Isolation	4 Tests	**100% PASS**	Zero leakage between User A and User B. Tables strictly filtered by user ID.
Suite 2: Items CRUD & Archiving	5 Tests	**100% PASS**	Soft-archiving (archived_at) verified. Active queries exclude deleted items.
Suite 3: Reminders & SMTP Dispatch	2 Tests	**100% PASS**	Due reminder detection accurate. Retry fallback increments attempt counter.
Suite 4: Backup Export & Security	3 Tests	**100% PASS**	Full 10-table JSON export with schema versioning and clean 'My PA' branding.
Suite 5: Settings & Model Tuning	2 Tests	**100% PASS**	Model selection and temperature/top-p sliders persist seamlessly.

Suite 6: Chat & Skills Execution	4 Tests	**100% PASS**	/info cheat sheet and /today execute with zero residual legacy branding.
Cognitive Domain Benchmark	4 Tests	**100% PASS**	100% precision distinguishing *Work* vs *Learning* and *Groceries* vs *Expenses*.

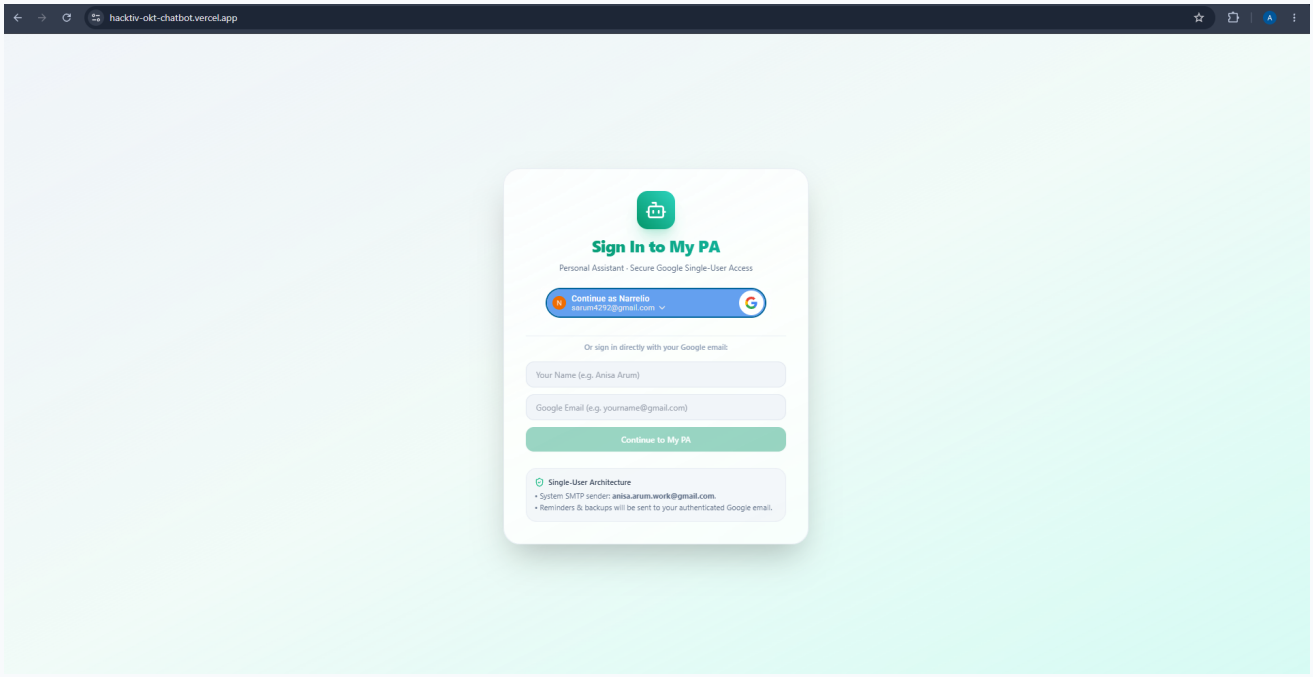
5. Application Visual Walkthrough & Interface Captures

Complete interface walkthrough featuring real production captures across both **Desktop Web (16:9 Landscape)** and **Mobile Device / PWA (9:20 Portrait)** views from the live deployment (https://hacktiv-okt-chatbot.vercel.app). All screenshots strictly maintain their authentic native proportions without stretching:

5.1 Authentication & Login Portal

Glassmorphic landing page featuring Google OAuth 2.0 single-sign-on and profile security credentials.

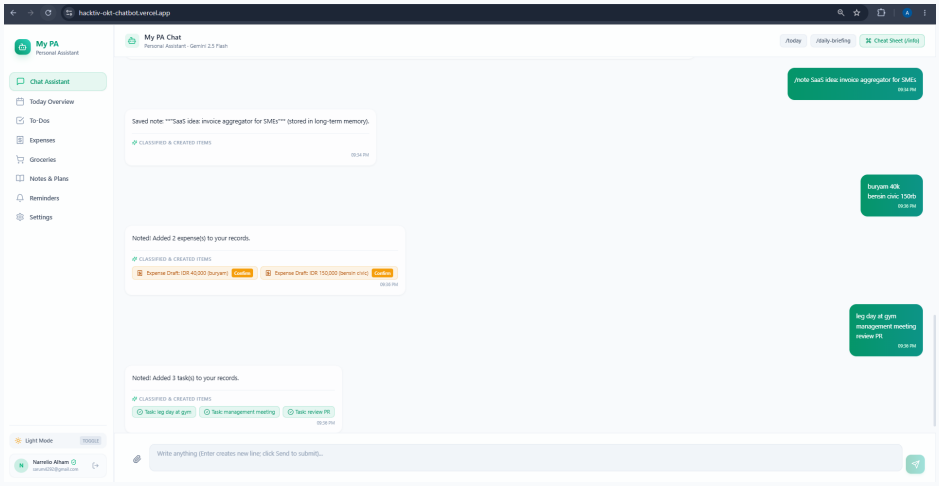
Desktop Web View



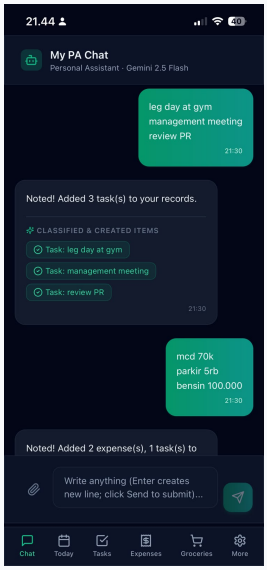
5.2 Stream-of-Consciousness Chat Interface

Conversational dialog view with natural language parsing, audio/media upload, and interactive skill execution.

■■ Desktop Web View (Landscape)



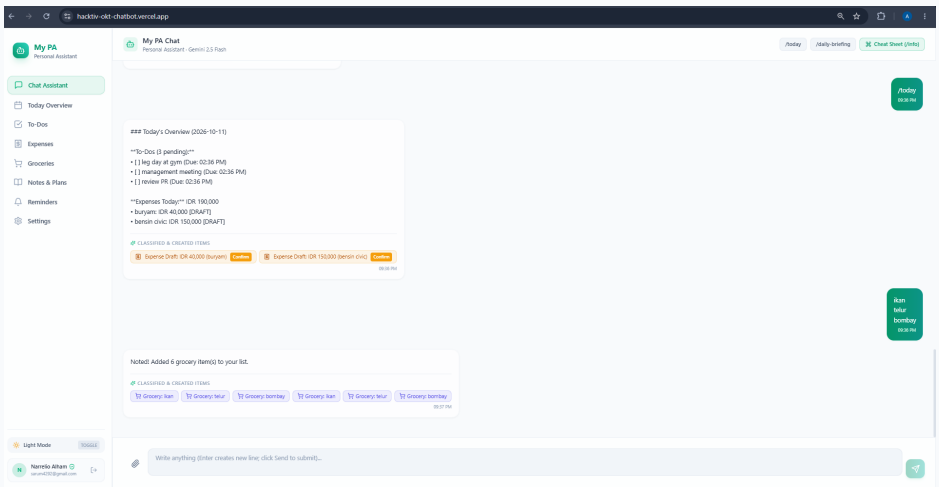
■ Mobile View (True Portrait)



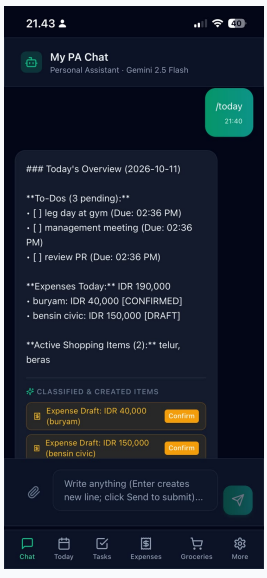
5.3 Multi-Turn Dialog & Memory Retrieval

Chat history demonstrating context recall from 3-tier memory and structured entity extraction feedback.

■■ Desktop Web View (Landscape)



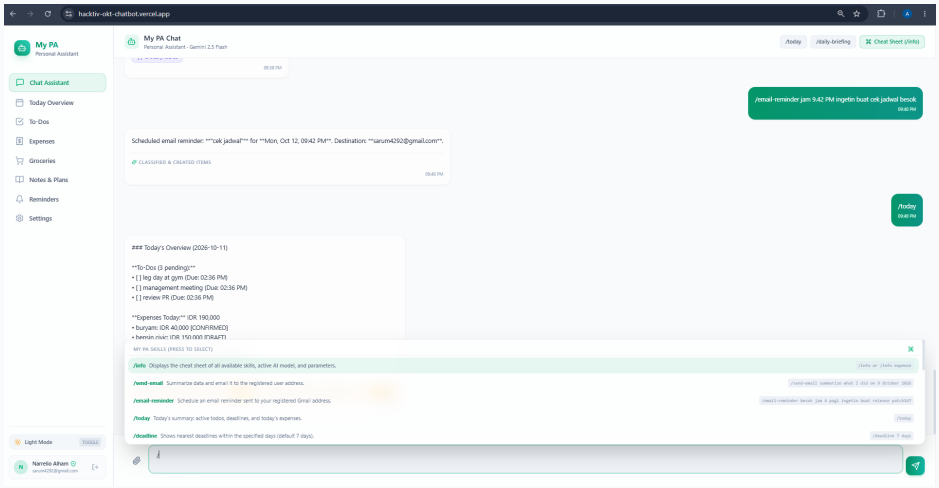
■ Mobile View (True Portrait)



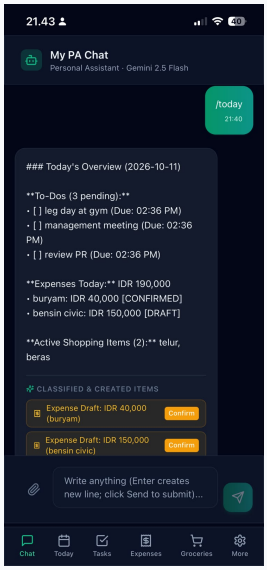
5.4 Slash Skills Dropdown & Auto-Suggestions

Real-time autocomplete menu triggered on keyboard '/' or header click for instant zero-quota skills.

■■ Desktop Web View (Landscape)



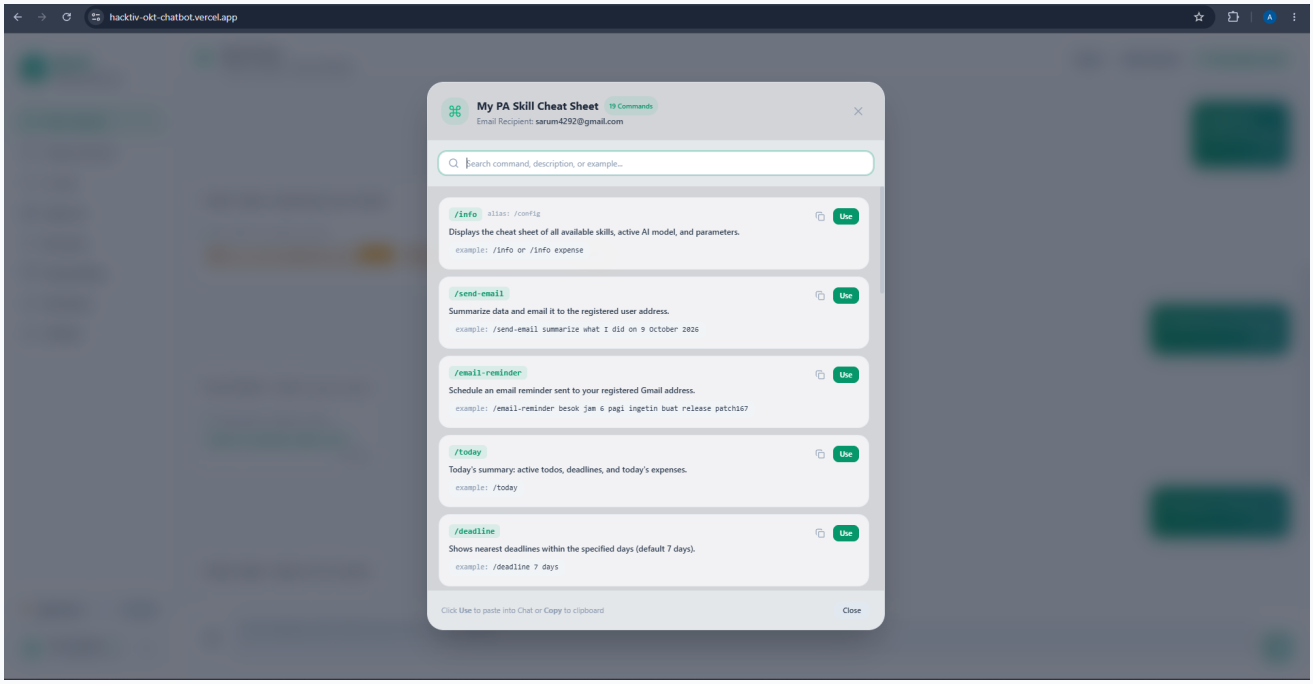
■ Mobile View (True Portrait)



5.5 Instant Skill Cheat Sheet Modal (/info)

Comprehensive skills directory modal displaying all 19 commands and active AI model parameters.

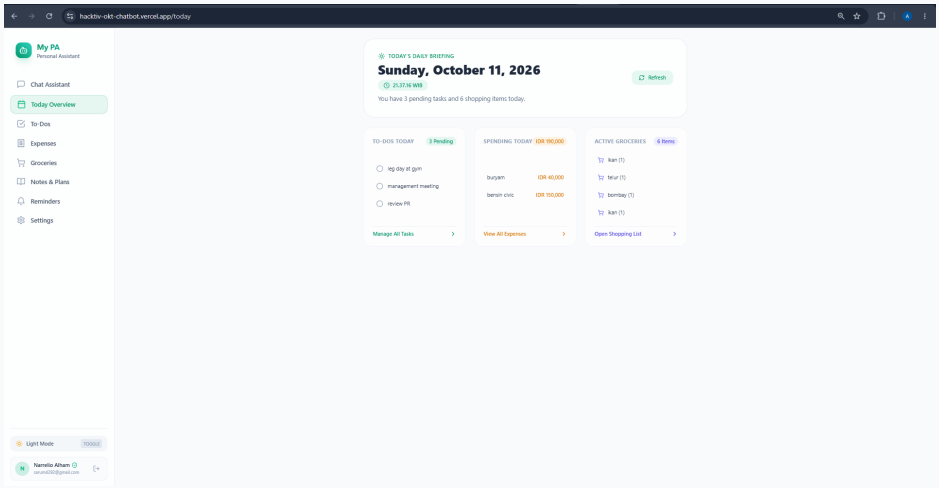
■■ Desktop Web View



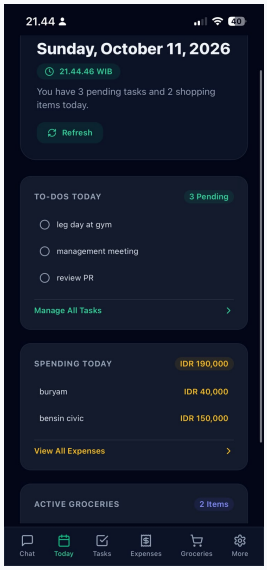
5.6 Daily Agenda & 'Today' Overview Dashboard

Consolidated productivity dashboard aggregating today's active tasks, pending deadlines, and daily expenses tally.

■■ Desktop Web View (Landscape)



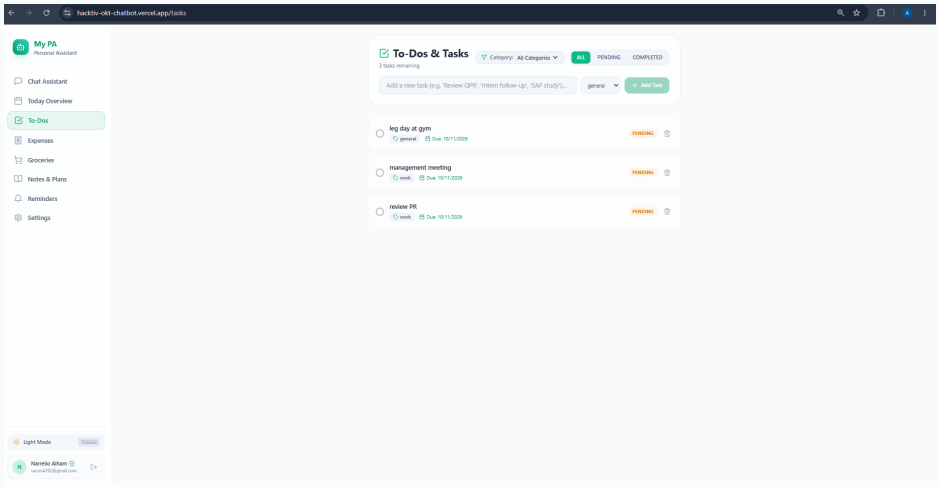
■ Mobile View (True Portrait)



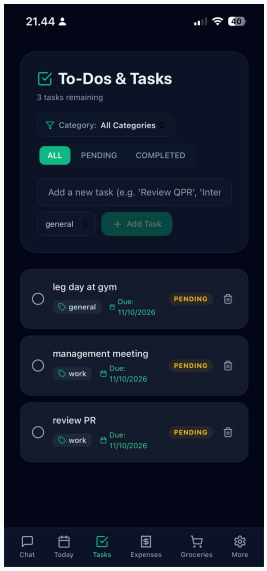
5.7 Tasks & To-Do Management

Interactive task tracker with category tags (work, finance, personal), due dates, and completion status toggles.

■■ Desktop Web View (Landscape)



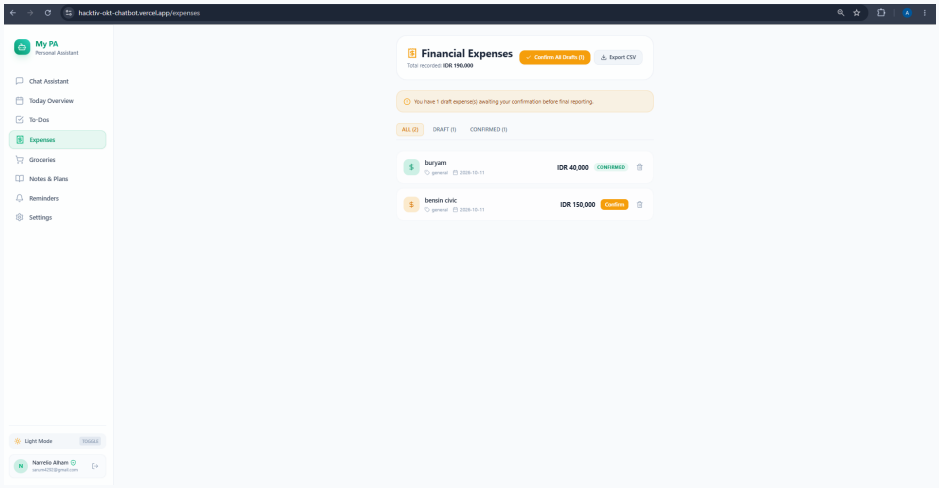
■ Mobile View (True Portrait)



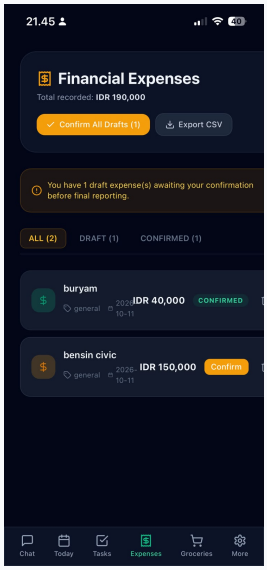
5.8 Financial Expenses Ledger & CSV Export

Financial expenditure tracker featuring DRAFT confirmation, category breakdowns, and 1-click CSV download.

■■ Desktop Web View (Landscape)



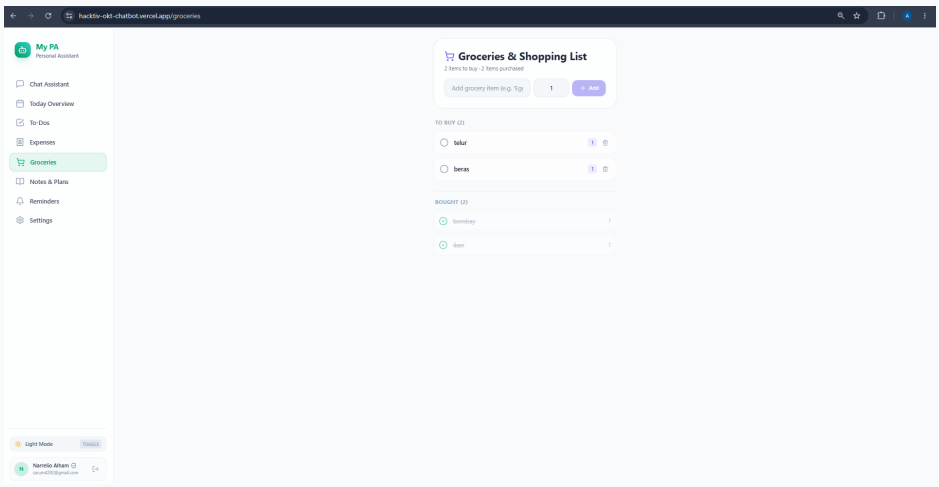
■ Mobile View (True Portrait)



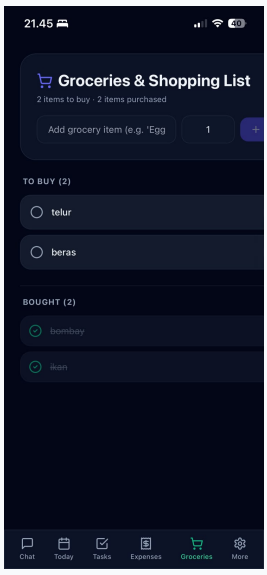
5.9 Interactive Groceries Shopping Checklist

Smart shopping list automatically populated from food items without explicit prices, featuring instant check-off.

■■ Desktop Web View (Landscape)



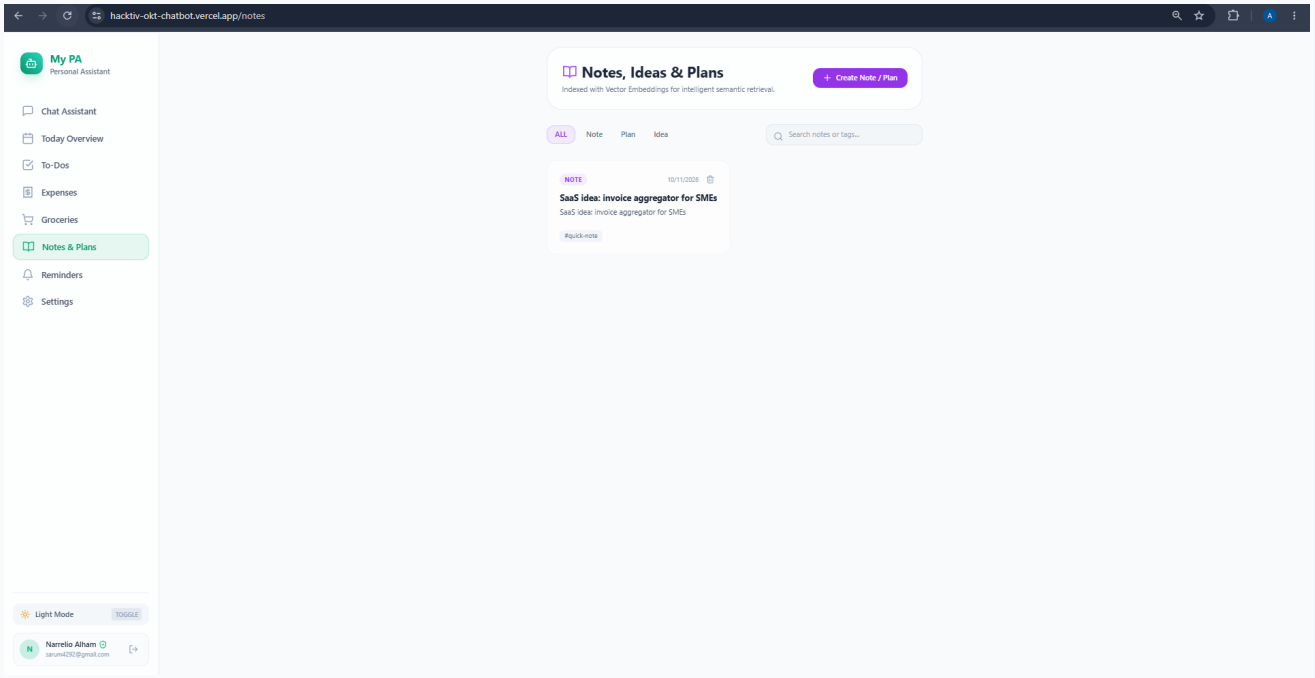
■ Mobile View (True Portrait)



5.10 Notes, Plans & Vector Knowledge Base

Long-term knowledge storage featuring markdown formatting, automated tag clustering, and 768-dim embeddings.

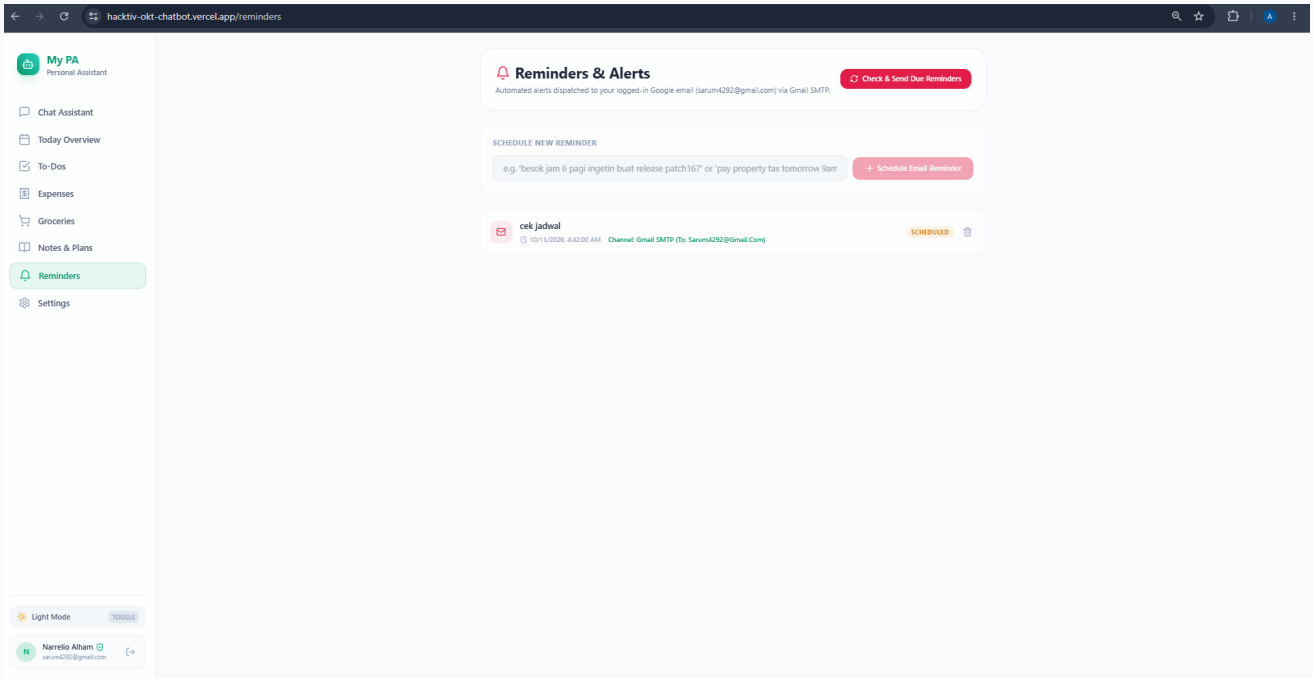
Desktop Web View



5.11 Reminders & Automated SMTP Dispatcher

Notification scheduling view tracking status (SCHEDULED, SENT, FAILED) across Gmail SMTP and in-app channels.

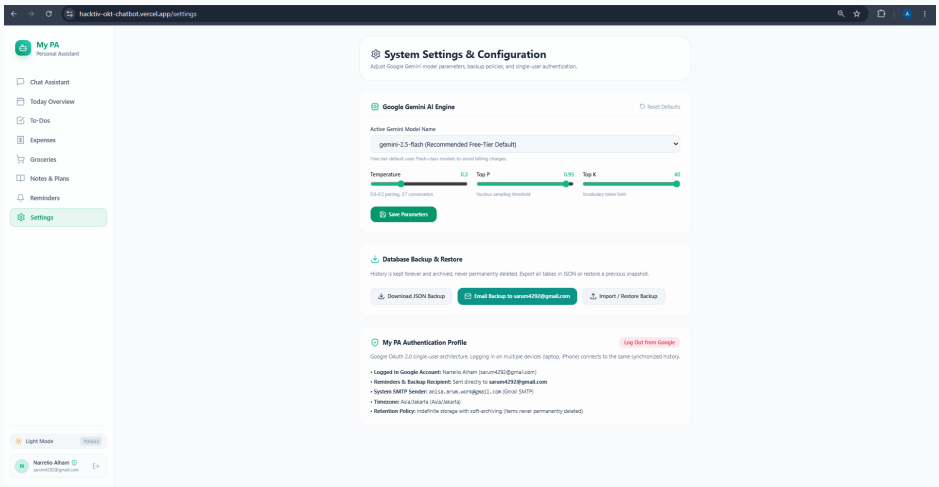
Desktop Web View



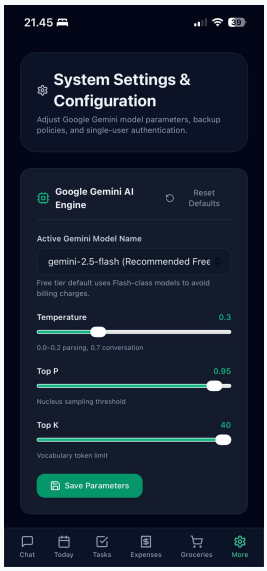
5.12 AI Model Tuning, Sliders & Database Backup

System settings view with AI hyperparameter sliders (Temperature, Top-P, Top-K sliders), theme switcher, and JSON backup.

■■ Desktop Web View (Landscape)

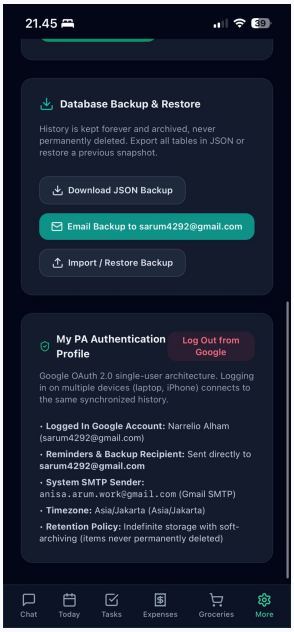


■ Mobile View (True Portrait)



5.13 Mobile Settings & Theme Customization View

Detailed mobile view of full-screen model settings, data retention policies, and export controls on handheld devices.



6. Project Summary & Reviewer Checklist

My PA fulfills all requirements established in the PRD. The application successfully decouples user input from rigid category menus, handles natural language ambiguity via Gemini 2.5 Flash, guarantees data persistence via Neon PostgreSQL + pgvector, and delivers email reminders via Gmail SMTP across both desktop and mobile platforms.

Feature / Criterion	Implementation Status
Stream-of-Consciousness Parsing	■ Implemented & 100% verified (Gemini Structured Outputs)
19 Built-in Skills (Slash Commands)	■ Implemented with autocomplete & 0ms instant replies
Multi-User Scoping & Isolation	■ Google OAuth authenticated with user-scoped database queries
Neon Serverless PostgreSQL + pgvector	■ 10 tables migrated with 768-dim vector embeddings
Automated Gmail SMTP Reminders	■ Background runner + REST endpoint dispatch
1-Click JSON Backup & Import	■ One-click download & instant email backup dispatch
Responsive Dual Layout (Web + Mobile PWA)	■ Fully optimized for desktop browsers and mobile touchscreens
Live Public Deployment	■ Accessible at https://hacktiv-okt-chatbot.vercel.app/